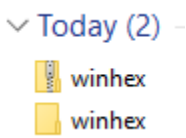
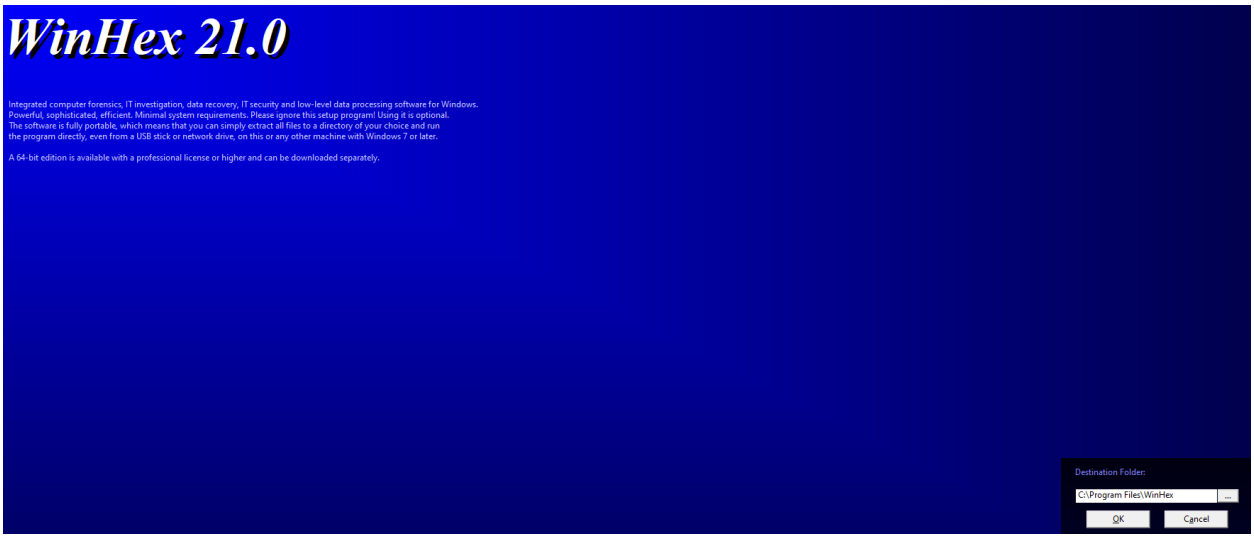
Navid Baghaei

**Lab 5**

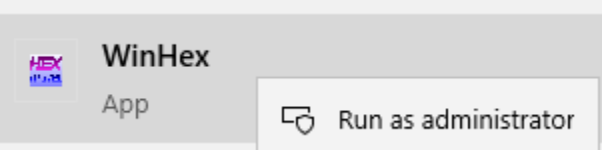
**TASK 1**

1. 

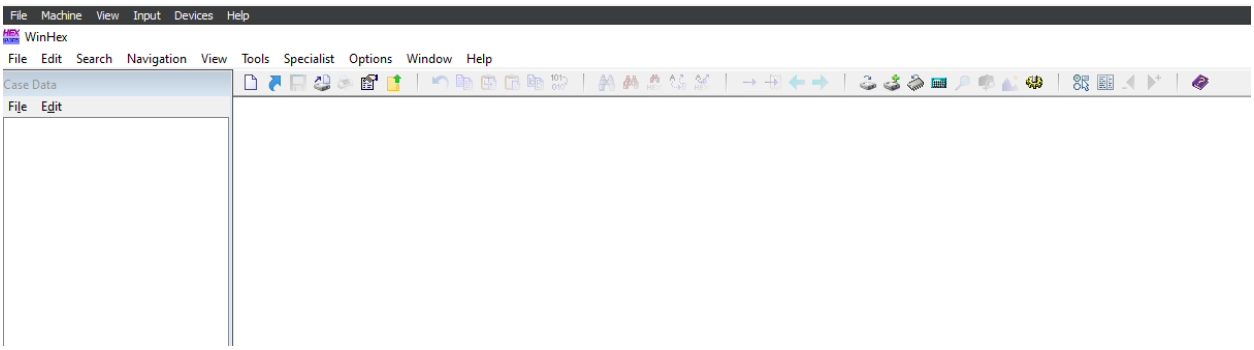

Downloaded and extracted winhex.



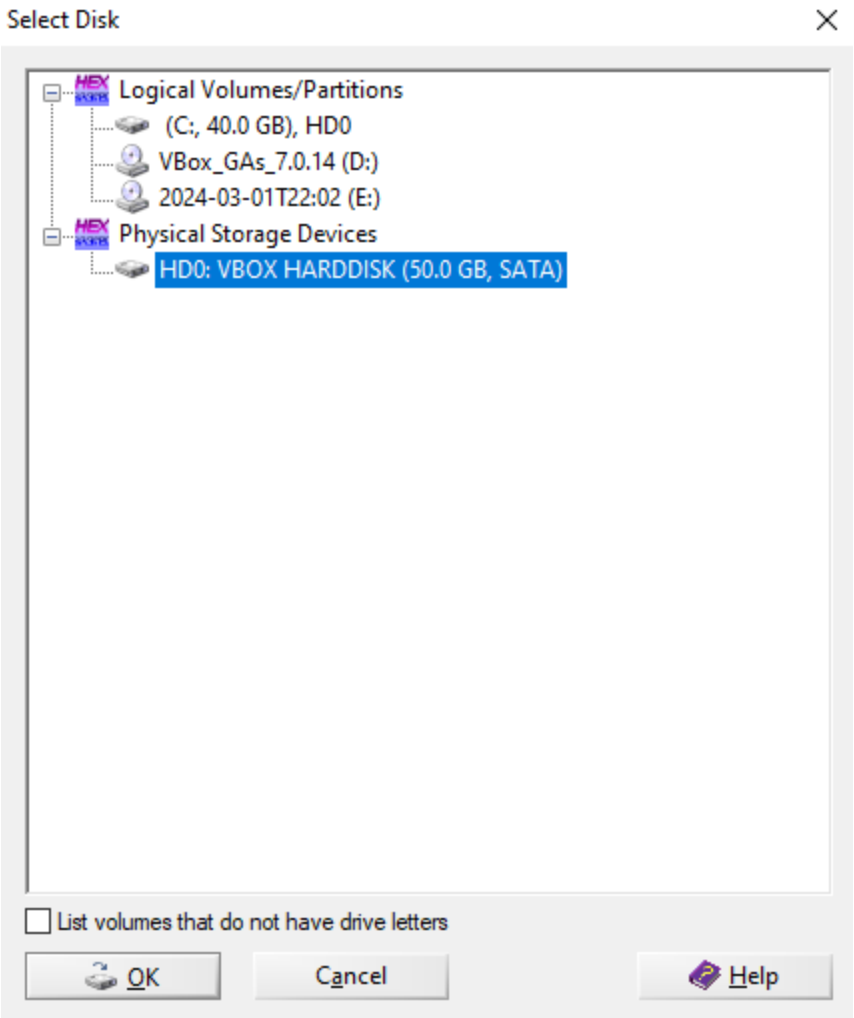

Ran the setup process and installed WinHex.



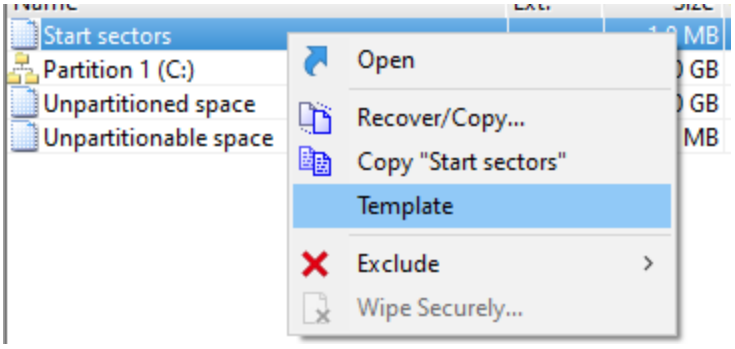

Ran the program as an administrator.




It launches with no issues.

2. 


Went to Tools -> Open Disk, and selected the HD0: VBOX HARDDISK option.

3. 

Master Boot Record, Base Offset: 0

| Offset | Title | Value |
|---|---|---|
| 0 | Master bootstrap loader | 33 C0 8E D0 BC 00 7C 8E C0 8E D8 BE 00 7 |
| 1B8 | Windows disk signature | B1 D3 7B 89 |
| 1B8 | Same reversed | 897BD3B1 |

Partition Table Entry #1

| | | |
|---|---|---|
| 1BE | 80 = active partition | 80 |
| 1BF | Start head | 32 |
| 1C0 | Start sector | 33 |
| 1C0 | Start cylinder | 0 |
| 1C2 | Partition type indicator ( | 07 |
| 1C3 | End head | 254 |
| 1C4 | End sector | 63 |
| 1C4 | End cylinder | 1,023 |
| 1C6 | Sectors preceding partiti | 2,048 |
| 1CA | Sectors in partition 1 | 83,881,984 |

Partition Table Entry #2

| | | |
|---|---|---|
| 1CE | 80 = active partition | 0 |

Right clicked on start sectors, and selected Template. It launched the Master Boot Record page, which shows the info above.

Go To Offset

New position: 1BE Bytes (hexadec.)

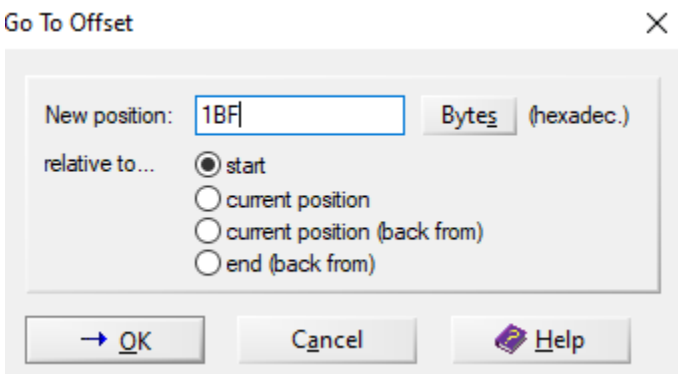
relative to...
- start
- current position
- current position (back from)
- end (back from)

OK Cancel Help

Clicked Navigation -> Go To Offset, then entered “1BE” as the position, and pressed OK.

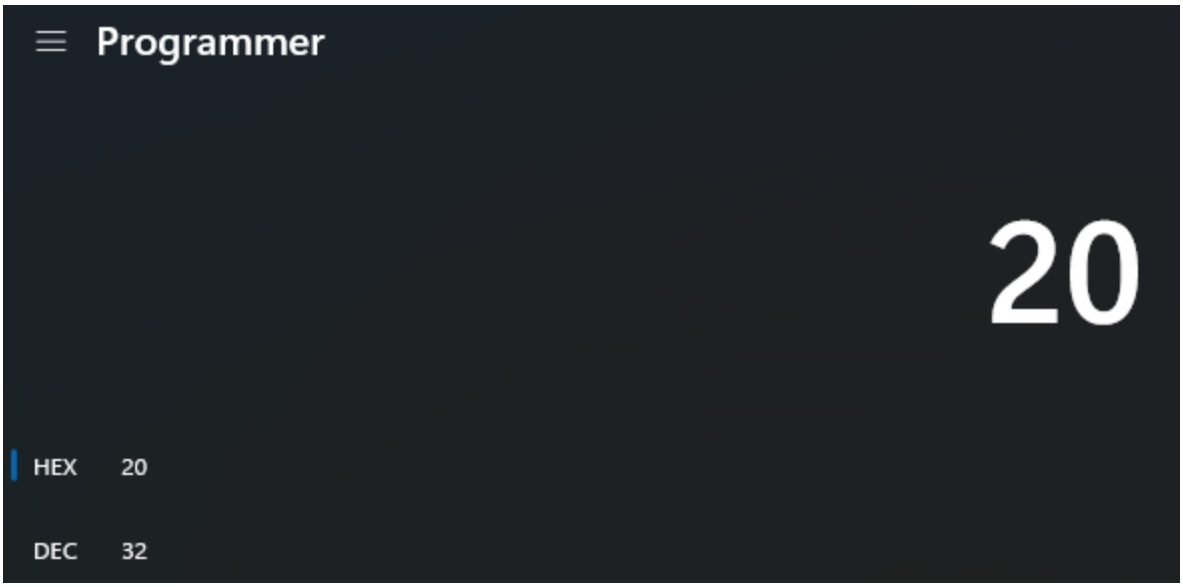
| Offset | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | A | B | C | D | E | F | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 1A | 1B | 1C | 1D | 1E | 1F |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 000000000 | 33 | C0 | 8E | D0 | BC | 00 | 7C | 8E | C0 | 8E | D8 | BE | 00 | 7C | BF | 00 | 06 | B9 | 00 | 02 | FC | F3 | A4 | 50 | 68 | 1C | 06 | CB | FB | B9 | 04 | 00 |
| 000000020 | BD | BE | 07 | 80 | 7E | 00 | 00 | 7C | 0B | 0F | 85 | 0E | 01 | 83 | C5 | 10 | E2 | F1 | CD | 18 | 88 | 56 | 00 | 55 | C6 | 46 | 11 | 05 | C6 | 46 | 10 | 00 |
| 000000040 | B4 | 41 | BB | AA | 55 | CD | 13 | 5D | 72 | 0F | 81 | FB | 55 | AA | 75 | 09 | F7 | C1 | 01 | 00 | 74 | 03 | FE | 46 | 10 | 66 | 60 | 80 | 7E | 10 | 00 | 74 |
| 000000060 | 26 | 66 | 68 | 00 | 00 | 00 | 00 | 66 | FF | 76 | 08 | 68 | 00 | 00 | 68 | 00 | 7C | 68 | 01 | 00 | 68 | 10 | 00 | B4 | 42 | 8A | 56 | 00 | 8B | F4 | CD | 13 |
| 000000080 | 9F | 83 | C4 | 10 | 9E | EB | 14 | B8 | 01 | 02 | BB | 00 | 7C | 8A | 56 | 00 | 8A | 76 | 01 | 8A | 4E | 02 | 8A | 6E | 03 | CD | 13 | 66 | 61 | 73 | 1C | FE |
| 0000000A0 | 4E | 11 | 75 | 0C | 80 | 7E | 00 | 80 | 0F | 84 | 8A | 00 | B2 | 80 | EB | 84 | 55 | 32 | E4 | 8A | 56 | 00 | CD | 13 | 5D | EB | 9E | 81 | 3E | FE | 7D | 55 |
| 0000000C0 | AA | 75 | 6E | FF | 76 | 00 | E8 | 8D | 00 | 75 | 17 | FA | B0 | D1 | E6 | 64 | E8 | 83 | 00 | B0 | DF | E6 | 60 | E8 | 7C | 00 | B0 | FF | E6 | 64 | E8 | 75 |
| 0000000E0 | 00 | FB | B8 | 00 | BB | CD | 1A | 66 | 23 | C0 | 75 | 3B | 66 | 81 | FB | 54 | 43 | 50 | 41 | 75 | 32 | 81 | F9 | 02 | 01 | 72 | 2C | 66 | 68 | 07 | BB | 00 |
| 000000100 | 00 | 66 | 68 | 00 | 02 | 00 | 00 | 66 | 68 | 08 | 00 | 00 | 00 | 66 | 53 | 66 | 53 | 66 | 55 | 66 | 68 | 00 | 00 | 00 | 00 | 66 | 68 | 00 | 7C | 00 | 00 | 66 |
| 000000120 | 61 | 68 | 00 | 00 | 07 | CD | 1A | 5A | 32 | F6 | EA | 00 | 7C | 00 | 00 | CD | 18 | A0 | B7 | 07 | EB | 08 | A0 | B6 | 07 | EB | 03 | A0 | B5 | 07 | 32 | E4 |
| 000000140 | 05 | 00 | 07 | 8B | F0 | AC | 3C | 00 | 74 | 09 | BB | 07 | 00 | B4 | 0E | CD | 10 | EB | F2 | F4 | EB | FD | 2B | C9 | E4 | 64 | EB | 00 | 24 | 02 | E0 | F8 |
| 000000160 | 24 | 02 | C3 | 49 | 6E | 76 | 61 | 6C | 69 | 64 | 20 | 70 | 61 | 72 | 74 | 69 | 74 | 69 | 6F | 6E | 20 | 74 | 61 | 62 | 6C | 65 | 00 | 45 | 72 | 72 | 6F | 72 |
| 000000180 | 20 | 6C | 6F | 61 | 64 | 69 | 6E | 67 | 20 | 6F | 70 | 65 | 72 | 61 | 74 | 69 | 6E | 67 | 20 | 73 | 79 | 73 | 74 | 65 | 6D | 00 | 4D | 69 | 73 | 73 | 69 | 6E |
| 0000001A0 | 67 | 20 | 6F | 70 | 65 | 72 | 61 | 74 | 69 | 6E | 67 | 20 | 73 | 79 | 73 | 74 | 65 | 6D | 00 | 00 | 00 | 63 | 7B | 9A | B1 | D3 | 7B | 89 | 00 | 00 | 80 | 20 |
| 0000001C0 | 21 | 00 | 07 | FE | FF | FF | 00 | 08 | 00 | 00 | 00 | F0 | FF | 04 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 |
| 0000001E0 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 55 | AA |

The cursor got set to position 1BE, which has the value of 80.



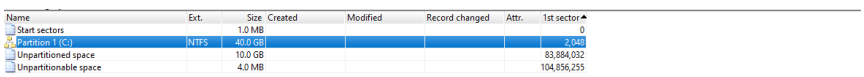

| Offset | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | A | B | C | D | E | F | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 1A | 1B | 1C | 1D | 1E | 1F |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 000000000 | 33 | C0 | 8E | D0 | BC | 00 | 7C | 8E | C0 | 8E | D8 | BE | 00 | 7C | BF | 00 | 06 | B9 | 00 | 02 | FC | F3 | A4 | 50 | 68 | 1C | 06 | CB | FB | B9 | 04 | 00 |
| 000000020 | BD | BE | 07 | 80 | 7E | 00 | 00 | 7C | 0B | 0F | 85 | 0E | 01 | 83 | C5 | 10 | E2 | F1 | CD | 18 | 88 | 56 | 00 | 55 | C6 | 46 | 11 | 05 | C6 | 46 | 10 | 00 |
| 000000040 | B4 | 41 | BB | AA | 55 | CD | 13 | 5D | 72 | 0F | 81 | FB | 55 | AA | 75 | 09 | F7 | C1 | 01 | 00 | 74 | 03 | FE | 46 | 10 | 66 | 60 | 80 | 7E | 10 | 00 | 74 |
| 000000060 | 26 | 66 | 68 | 00 | 00 | 00 | 00 | 66 | FF | 76 | 08 | 68 | 00 | 00 | 68 | 00 | 7C | 68 | 01 | 00 | 68 | 10 | 00 | B4 | 42 | 8A | 56 | 00 | 8B | F4 | CD | 13 |
| 000000080 | 9F | 83 | C4 | 10 | 9E | EB | 14 | B8 | 01 | 02 | BB | 00 | 7C | 8A | 56 | 00 | 8A | 76 | 01 | 8A | 4E | 02 | 8A | 6E | 03 | CD | 13 | 66 | 61 | 73 | 1C | FE |
| 0000000A0 | 4E | 11 | 75 | 0C | 80 | 7E | 00 | 80 | 0F | 84 | 8A | 00 | B2 | 80 | EB | 84 | 55 | 32 | E4 | 8A | 56 | 00 | CD | 13 | 5D | EB | 9E | 81 | 3E | FE | 7D | 55 |
| 0000000C0 | AA | 75 | 6E | FF | 76 | 00 | E8 | 8D | 00 | 75 | 17 | FA | B0 | D1 | E6 | 64 | E8 | 83 | 00 | B0 | DF | E6 | 60 | E8 | 7C | 00 | B0 | FF | E6 | 64 | E8 | 75 |
| 0000000E0 | 00 | FB | B8 | 00 | BB | CD | 1A | 66 | 23 | C0 | 75 | 3B | 66 | 81 | FB | 54 | 43 | 50 | 41 | 75 | 32 | 81 | F9 | 02 | 01 | 72 | 2C | 66 | 68 | 07 | BB | 00 |
| 000000100 | 00 | 66 | 68 | 00 | 02 | 00 | 00 | 66 | 68 | 08 | 00 | 00 | 00 | 66 | 53 | 66 | 53 | 66 | 55 | 66 | 68 | 00 | 00 | 00 | 00 | 66 | 68 | 00 | 7C | 00 | 00 | 66 |
| 000000120 | 61 | 68 | 00 | 00 | 07 | CD | 1A | 5A | 32 | F6 | EA | 00 | 7C | 00 | 00 | CD | 18 | A0 | B7 | 07 | EB | 08 | A0 | B6 | 07 | EB | 03 | A0 | B5 | 07 | 32 | E4 |
| 000000140 | 05 | 00 | 07 | 8B | F0 | AC | 3C | 00 | 74 | 09 | BB | 07 | 00 | B4 | 0E | CD | 10 | EB | F2 | F4 | EB | FD | 2B | C9 | E4 | 64 | EB | 00 | 24 | 02 | E0 | F8 |
| 000000160 | 24 | 02 | C3 | 49 | 6E | 76 | 61 | 6C | 69 | 64 | 20 | 70 | 61 | 72 | 74 | 69 | 74 | 69 | 6F | 6E | 20 | 74 | 61 | 62 | 6C | 65 | 00 | 45 | 72 | 72 | 6F | 72 |
| 000000180 | 20 | 6C | 6F | 61 | 64 | 69 | 6E | 67 | 20 | 6F | 70 | 65 | 72 | 61 | 74 | 69 | 6E | 67 | 20 | 73 | 79 | 73 | 74 | 65 | 6D | 00 | 4D | 69 | 73 | 73 | 69 | 6E |
| 0000001A0 | 67 | 20 | 6F | 70 | 65 | 72 | 61 | 74 | 69 | 6E | 67 | 20 | 73 | 79 | 73 | 74 | 65 | 6D | 00 | 00 | 00 | 63 | 7B | 9A | B1 | D3 | 7B | 89 | 00 | 00 | 80 | 20 |
| 0000001C0 | 21 | 00 | 07 | FE | FF | FF | 00 | 08 | 00 | 00 | 00 | F0 | FF | 04 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 | 00 |

Using Go To Offset again, and entering 1BF takes us to the value of 20. The lab mentions how this is because the template uses decimal values while this shows hexadecimal values.

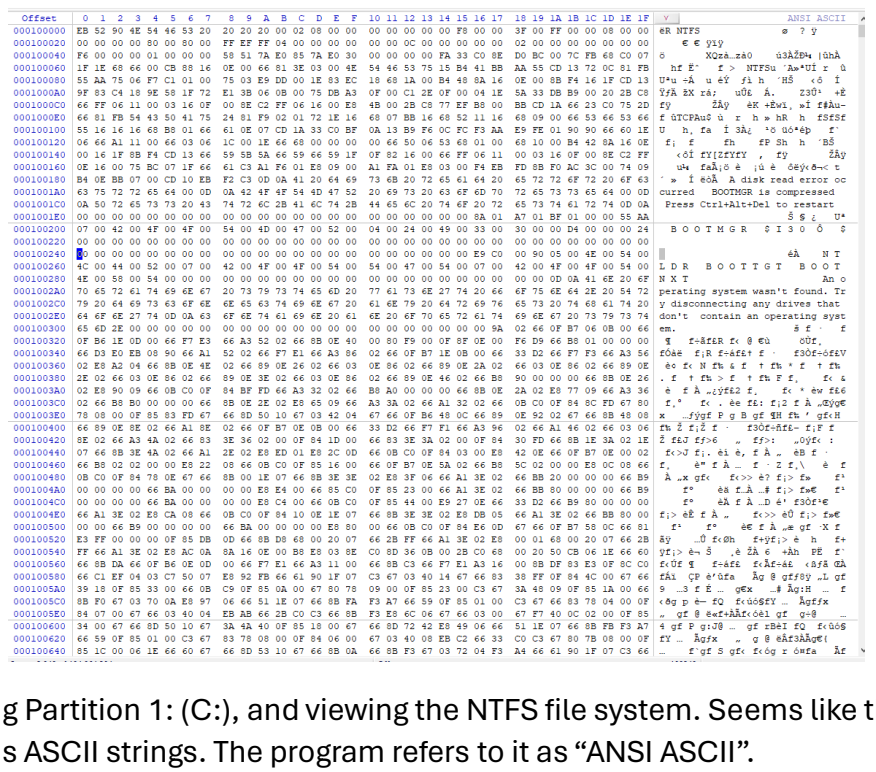



20 in hexadecimal converted to decimal is equal to 32, which is what was shown on the template.

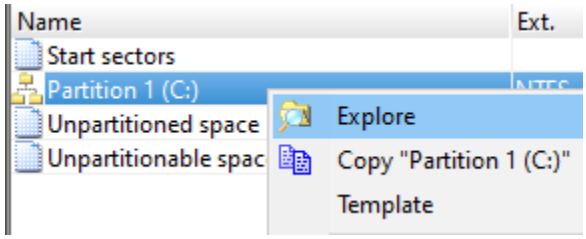
4.



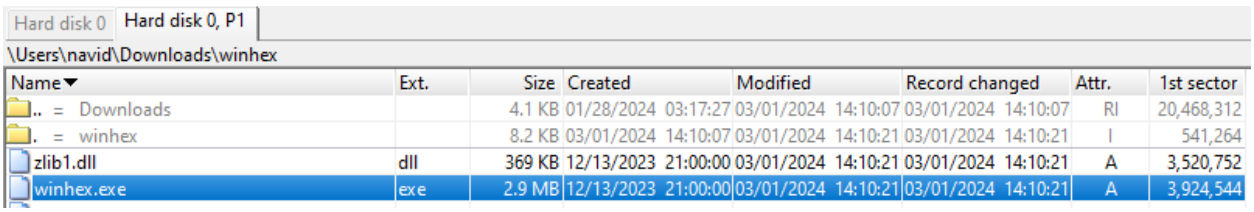
| Name | Ext. | Size | Created | Modified | Record changed | Attr. | 1st sector |
|---|---|---|---|---|---|---|---|
| Start sectors | | 1.0 MB | | | | | 0 |
| Partition 1 (C:) | NTFS | 40.0 GB | | | | | 2,048 |
| Unpartitioned space | | 10.0 GB | | | | | 83,884,032 |
| Unpartitionable space | | 4.0 MB | | | | | 104,856,255 |



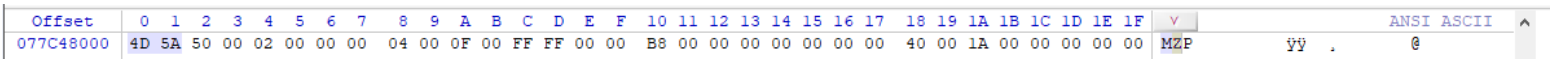
Selecting Partition 1: (C:), and viewing the NTFS file system. Seems like the information is stored as ASCII strings. The program refers to it as “ANSI ASCII”.

5. 


Right clicked on Partition 1 (C:), then clicked Explore.




Navigated to the downloads folder where winhex was downloaded, and selected winhex.exe.




Selected the first two bytes, which highlighted the ASCII representation on the right side. “MZ”.

| | | | | |
|---|---|---|---|---|
| 4D 5A | MZ | 0 | exe<br>dll<br>mui<br>sys<br>scr<br>cpl<br>ocx<br>ax<br>iec<br>ime<br>rs<br>tsp<br>fon<br>efi | DOS MZ executable and its descendants (including NE and PE) |

Following the Wikipedia link, and looking up 4D 5A shows that the byte set represents DOS MZ executable files and its descendants: exe, dll, mui, sys, scr, and more. Pretty cool.

6. winhex.chm

I decided to look at the .chm file.

| Offset | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | A | B | C | D | E | F | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 1A | 1B | 1C | 1D | 1E | 1F | V |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 077BA2000 | 49 | 54 | 53 | 46 | 03 | 00 | 00 | 00 | 60 | 00 | 00 | 00 | 01 | 00 | 00 | 00 | 8A | B8 | 5D | 73 | 07 | 04 | 00 | 00 | 10 | FD | 01 | 7C | AA | 7B | D0 | 11 | ITSF |

The first two bytes of it in hex are “49 54”. The ASCII representation is ITSF.

| | | | | |
|---|---|---|---|---|
| 49 54 53 46 03 00 00 00 60 00 00 00 | ITSF ETX NUL NUL NUL ` NUL NUL NUL | 0 | chm | MS Windows HtmlHelp Data |

Here it is on Wikipedia. The description is “MS Windows Html Help Data”.

**TASK 2**

1.

Name

InCh05

InCh05

Downloaded the InCh05 zip file, extracted it, and ran the exe file to extract the image file.

2. New Case

Help

Chain of Custody | Custom Fields | Case Narrative

Basic Case Data | Case Categories | Offense & Custody Data | Description of Evidence

Case Name: InChap05

Case Type: Criminal

Investigator: Navid Baghaei

Organization:

Contact Details:

Timezone: Local (UTC -8:00) Pacific Time (US & Canada) — ☑ Account for Daylight Saving Time

Display Date Format: 3/2/2024 (Default) — ☐ Display timezone on dates

Default Drive: C:\ [Local]

Acquisition Type: ○ Live Acquisition of Current Machine ◉ Investigate Disk(s) from Another Machine

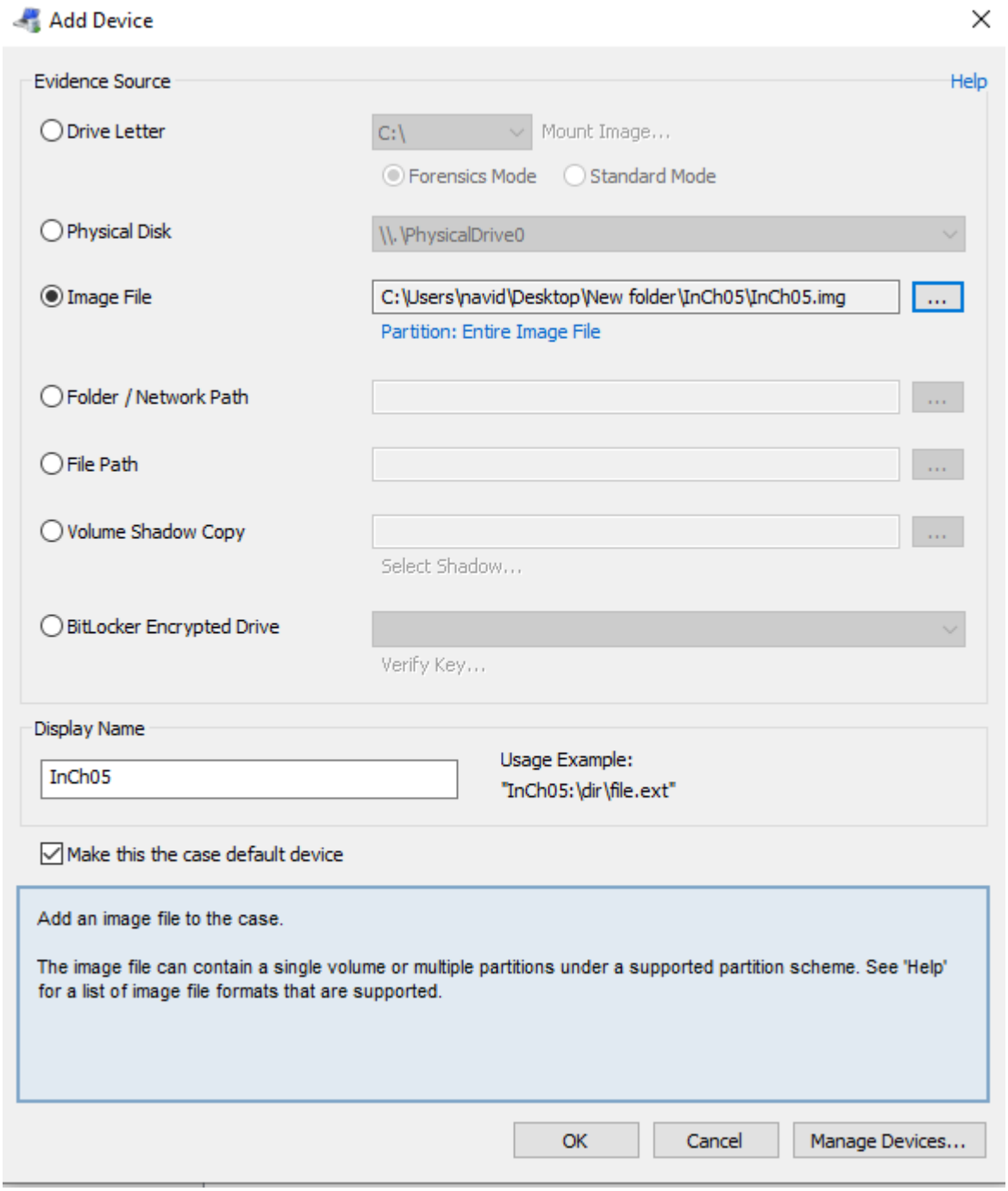
Case Folder: ◉ Default Location ○ Custom Location

C:\Users\navid\Documents\PassMark\OSForensics\Cases\InChap05\ Browse

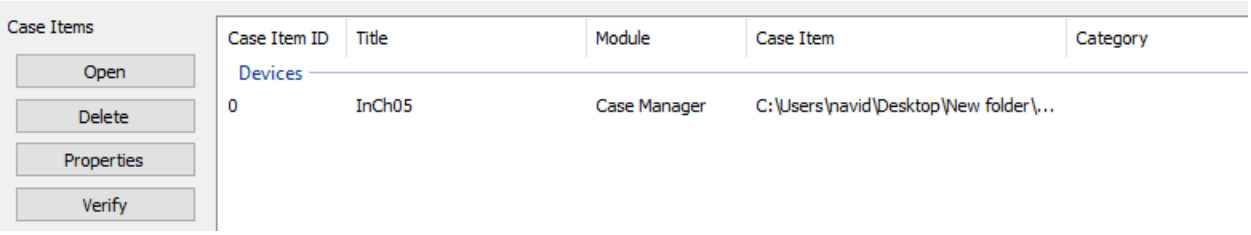
☑ Log case activity ☐ Enable USB Write-block

OK Cancel

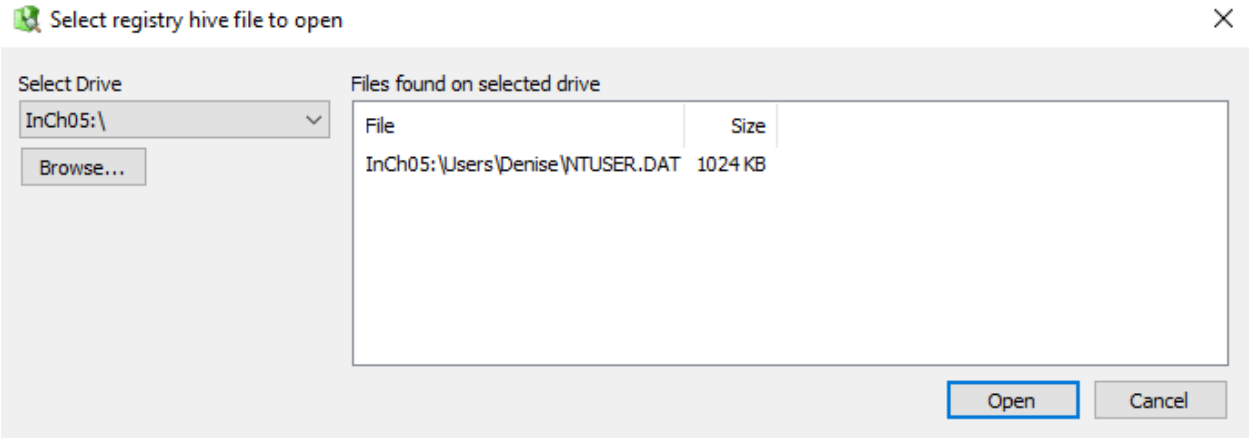
In OSForensics, I created a new case and named it “InChap05”. Set myself as the investigator and made sure “Investigate Disk(s) from Another Machine was selected.



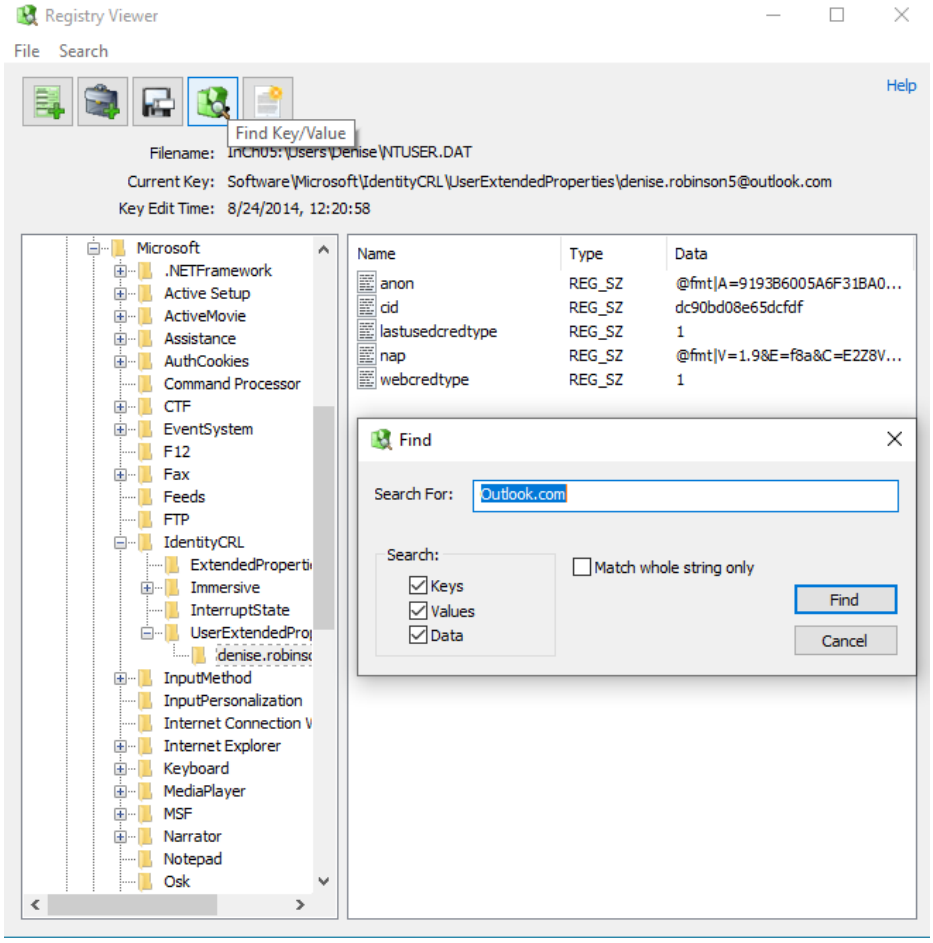

The “Add Device” menu automatically opened. I selected “Image File” and picked the image I extracted earlier in the InCh05 zip.



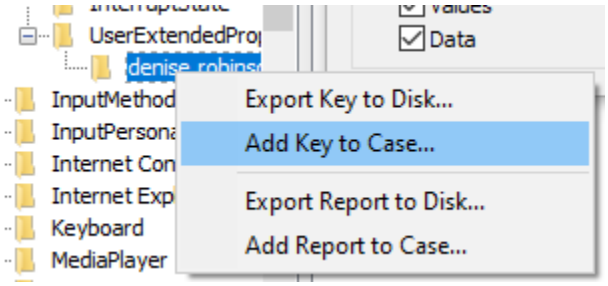

The img file now appears in the Case Items section.

3. 

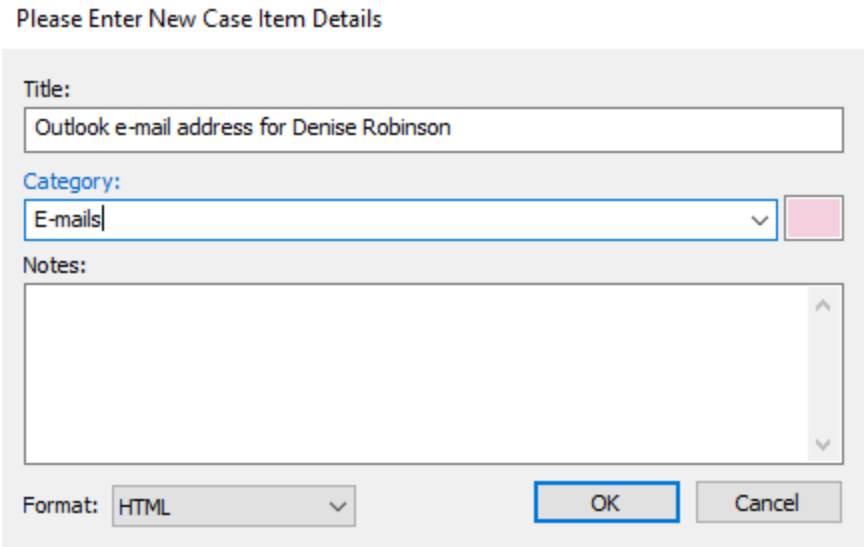

Clicked the Registry Viewer on the left tab, and it opened the menu above. The image file was auto mounted and a NTUSER.DAT file for the user Denise was automatically added. I selected it and pressed Open.



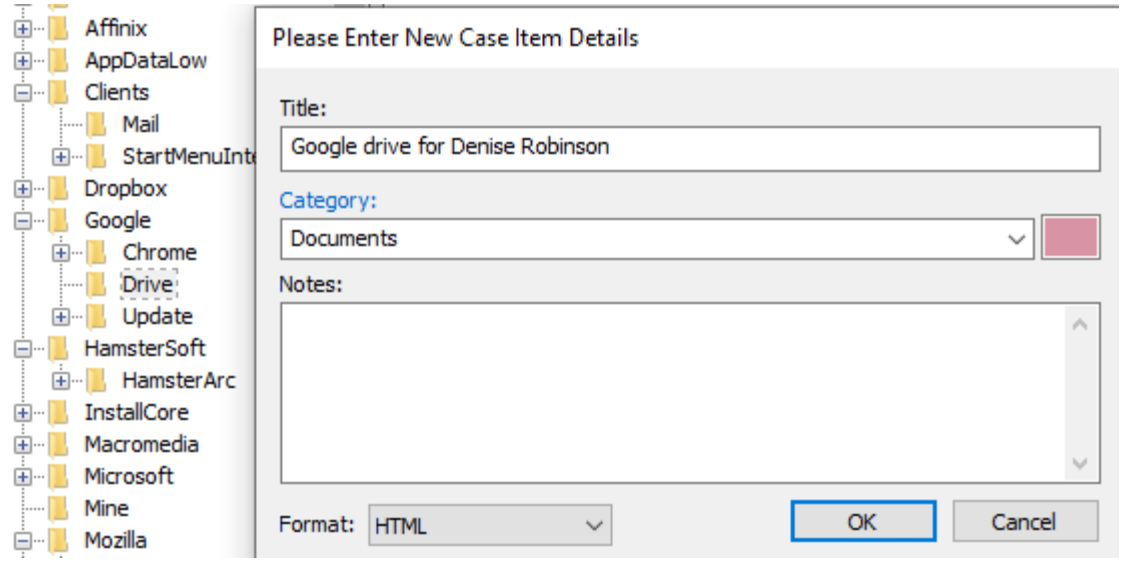
In the Registry Viewer, I selected “Find Key/Value” and searched for Outlook.com. The Registry tree expanded with a result.

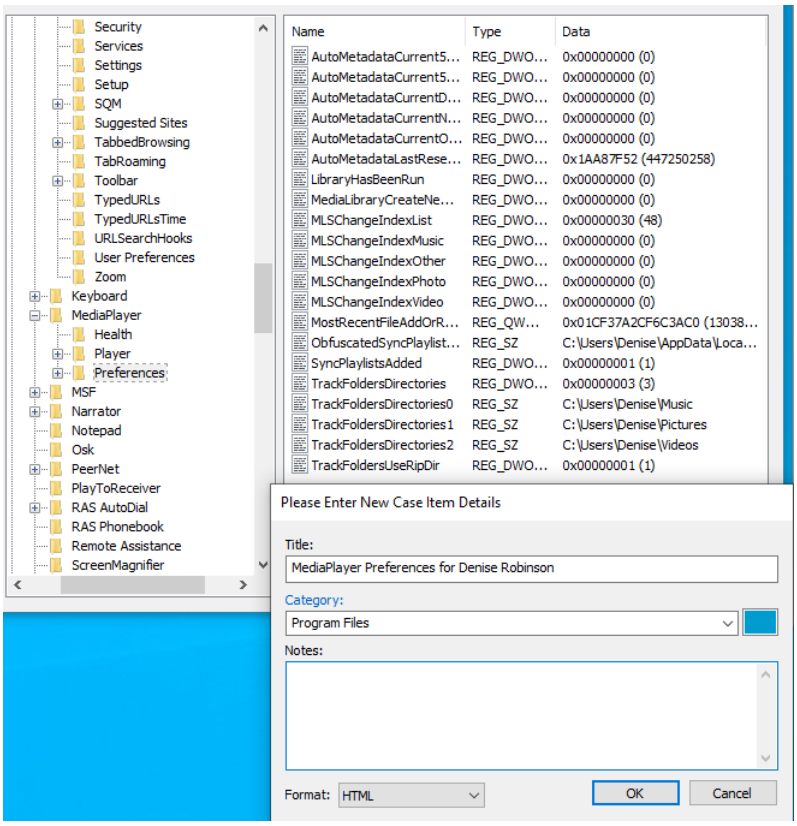


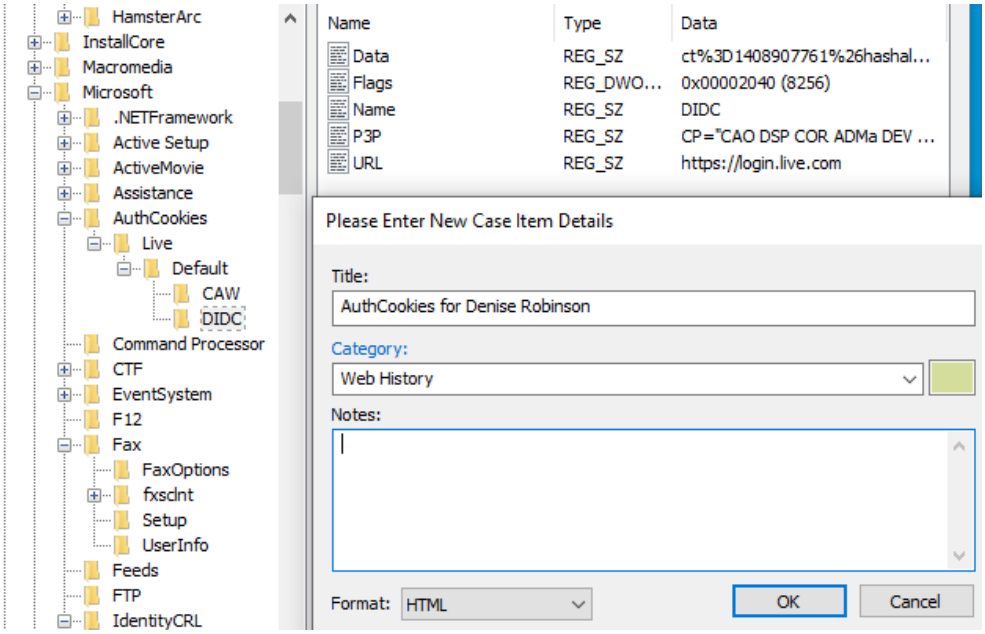

Right clicked Key and selected “Add Key to Case”




On the case item details menu, I titled it “Outlook e-mail address for Denise Robinson”. I also gave it the category of E-mails.

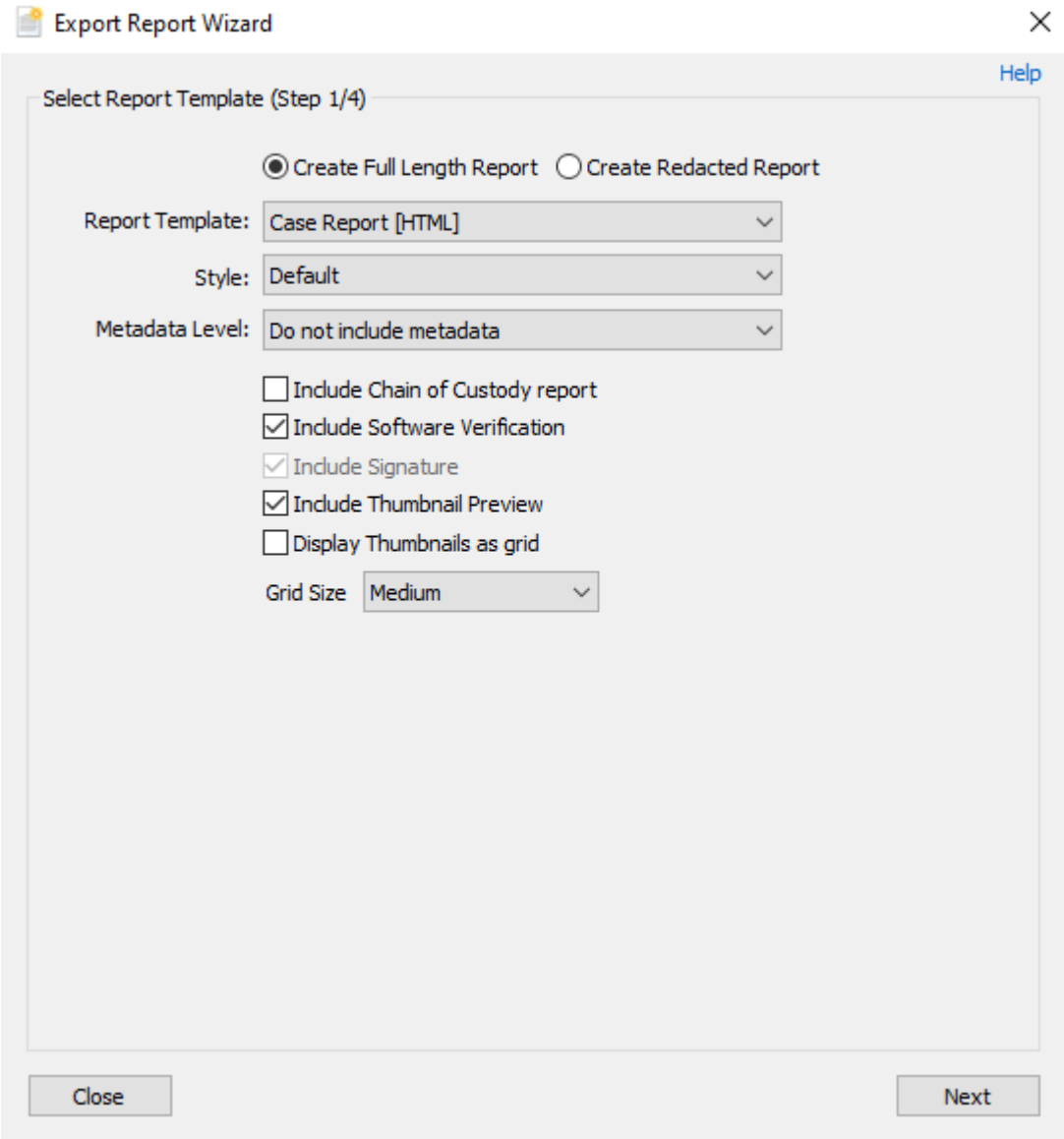

Affinix
AppDataLow
Clients
Mail
StartMenuInte
Dropbox
Google
Chrome
Drive
Update
HamsterSoft
HamsterArc
InstallCore
Macromedia
Microsoft
Mine
Mozilla
Please Enter New Case Item Details
Title:
Google drive for Denise Robinson
Category:
Documents
Notes:
Format: HTML
OK
Cancel


Security
Services
Settings
Setup
SQM
Suggested Sites
TabbedBrowsing
TabRoaming
Toolbar
TypedURLs
TypedURLsTime
URLSearchHooks
User Preferences
Zoom
Keyboard
MediaPlayer
Health
Player
Preferences
MSF
Narrator
Notepad
Osk
PeerNet
PlayToReceiver
RAS AutoDial
RAS Phonebook
Remote Assistance
ScreenMagnifier
Name
Type
Data
AutoMetadataCurrent5... REG_DWO... 0x00000000 (0)
AutoMetadataCurrent5... REG_DWO... 0x00000000 (0)
AutoMetadataCurrentD... REG_DWO... 0x00000000 (0)
AutoMetadataCurrentN... REG_DWO... 0x00000000 (0)
AutoMetadataCurrentO... REG_DWO... 0x00000000 (0)
AutoMetadataLastRese... REG_DWO... 0x1AA87F52 (447250258)
LibraryHasBeenRun REG_DWO... 0x00000000 (0)
MediaLibraryCreateNe... REG_DWO... 0x00000000 (0)
MLSChangeIndexList REG_DWO... 0x00000030 (48)
MLSChangeIndexMusic REG_DWO... 0x00000000 (0)
MLSChangeIndexOther REG_DWO... 0x00000000 (0)
MLSChangeIndexPhoto REG_DWO... 0x00000000 (0)
MLSChangeIndexVideo REG_DWO... 0x00000000 (0)
MostRecentFileAddOrR... REG_QW... 0x01CF37A2CF6C3AC0 (13038...
ObfuscatedSyncPlaylist... REG_SZ C:\Users\Denise\AppData\Loca...
SyncPlaylistsAdded REG_DWO... 0x00000001 (1)
TrackFoldersDirectories REG_DWO... 0x00000003 (3)
TrackFoldersDirectories0 REG_SZ C:\Users\Denise\Music
TrackFoldersDirectories1 REG_SZ C:\Users\Denise\Pictures
TrackFoldersDirectories2 REG_SZ C:\Users\Denise\Videos
TrackFoldersUseRipDir REG_DWO... 0x00000001 (1)
Please Enter New Case Item Details
Title:
MediaPlayer Preferences for Denise Robinson
Category:
Program Files
Notes:
Format: HTML
OK
Cancel

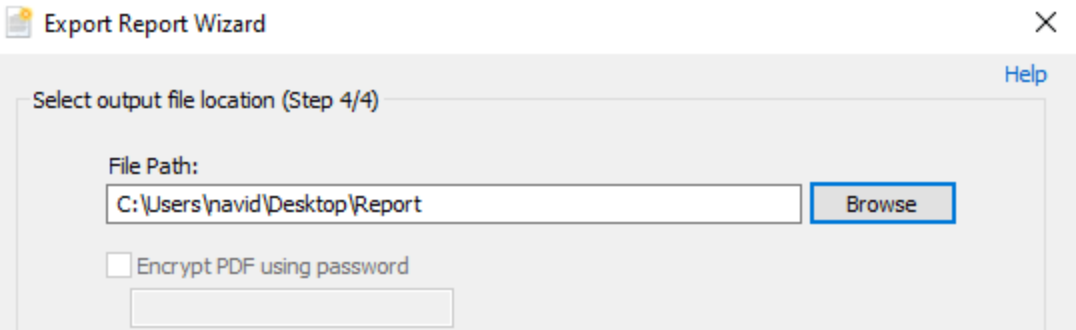

HamsterArc
InstallCore
Macromedia
Microsoft
.NETFramework
Active Setup
ActiveMovie
Assistance
AuthCookies
Live
Default
CAW
DIDC
Command Processor
CTF
EventSystem
F12
Fax
FaxOptions
fxsclnt
Setup
UserInfo
Feeds
FTP
IdentityCRL
Name
Type
Data
Data REG_SZ ct%3D1408907761%26hashal...
Flags REG_DWO... 0x00002040 (8256)
Name REG_SZ DIDC
P3P REG_SZ CP="CAO DSP COR ADMa DEV ...
URL REG_SZ https://login.live.com
Please Enter New Case Item Details
Title:
AuthCookies for Denise Robinson
Category:
Web History
Notes:
Format: HTML
OK
Cancel

| Case Item ID | Title | Module | Case Item | Category |
|---|---|---|---|---|
| Exported Items | | | | |
| 1 | Outlook e-mail address for... | Registry Viewer | RV 2024-03-02 01-22-37.html | E-mails |
| 2 | Google drive for Denise R... | Registry Viewer | RV 2024-03-02 01-28-54.html | Documents |
| 3 | AuthCookies for Denise R... | Registry Viewer | RV 2024-03-02 01-31-37.html | Web History |
| 4 | MediaPlayer Preferences f... | Registry Viewer | RV 2024-03-02 01-33-14.html | Program Files |
| Devices | | | | |
| 0 | InCh05 | Case Manager | C:\Users\navid\Desktop\New folder\... | |

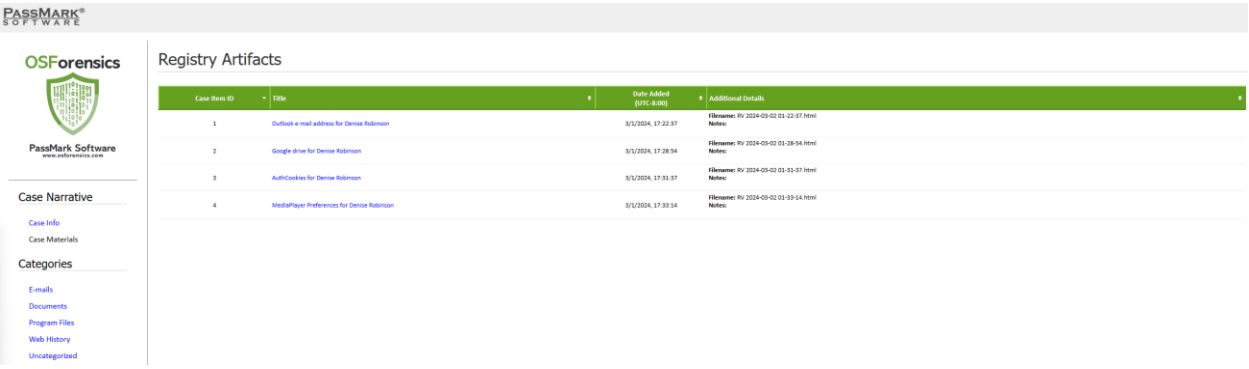
Added 3 more random keys and titled them accordingly. Closed the registry viewer.

4. 


On the Manage Case page, I clicked “Generate Report” which opened the Export Report Wizard.




Clicked next through the pages and gave the output file a path. Pressed OK and it started generating the report.




After it finished, I navigated to the folder where it was saved and opened the registry-artifacts file. It displays the 4 artifacts I added.

**Software\Microsoft\IdentityCRL\UserExtendedProperties\denise.robinson5@outlook.com**
File: InCh05:\Users\Denise\NTUSER.DAT
Timezone: UTC -8:00
Class name:
Last Write Time: 8/24/2014, 12:20:58

**Values**
Name: anon
Type: REG_SZ
Data: @fmt|A=9193B6005A6F31BA074C3927FFFFFFFF&E=fe4&W=1|bing.com;atdmt.com|Fri, 13-Mar-2015 02:20:58 GMT||false

Name: cid
Type: REG_SZ
Data: dc90bd08e65dcfdf

Name: lastusedcredtype
Type: REG_SZ
Data: 1

Name: nap
Type: REG_SZ
Data: @fmt|V=1.9&E=f8a&C=E2Z8VfmKFym4_nplO4tlprkr2pstYbV1XjtHRXxBlGZtOiFbsjJvqw&W=1|bing.com;atdmt.com|Wed, 03-Dec-2014 03:20:58 GMT||false

Name: webcredtype
Type: REG_SZ
Data: 1

Clicking on each one opens a view of all the data within that folder.